Contrastive Robustness for Genome Language Models

Cheng-You Ho **Hriday Meka**

1 Problem Definition

DNA is a string of nucleotides *A*, *C*, *T*, and *G*. Nucleotide triplets are called *codons*, most of which correspond to an amino acid, the fundamental element of proteins. There are 64 different codons; however, they only make 20 distinct types of amino acids. Because some codons make the same amino acid, it may be harmless to modify a nucleotide in a DNA sequence if the final protein turns out to be identical.

Genome Language Models (gLMs) such as DNABERT-2 (Zhou et al. [2024]) are transformers pretrained on DNA sequences. They can be fine-tuned for downstream tasks, such as classifying what DNA sequences lead to antibiotic-resistant proteins, and which do not.

Our goal for this project is to identify whether existing gLMs are robust to input nucleotide modifications in the task of classifying for antibacterial resistance. DNABERT-2 and the Nucleotide Transformer (Dalla-Torre et al. [2025]) will be our baselines. Because we know which codons have synonyms and which do not, we could cheaply create a variety of input sequences.

2 Relevance to Trustworthy Aspects

The main goal of this work is to *evaluate* the *robustness* of gLMs. A robust gLM should change its prediction only when the modification on the input is one what will change the final protein.

We will *extend* existing methods by investigating contrastive fine-tuning with synonymous and disruptive DNA variants. Finally, we seek to *explain* the gLM after fine-tuning to explore whether it is aligned with our intent.

3 Dataset

We will use two datasets: Comprehensive Antibiotic Resistance Database (CARD) and MEGARes. CARD is a collection of over 6,000 peer-reviewed resistance genes, their protein products, and phenotypes (e.g. resistance level.) MEGARes contains over 7,000 Nucleotide sequences spanning 4 antimicrobial types and over 50 resistance classes.

Summary statistics of the two datasets are shown in Figure 1, and a visualization of the dataset is shown in Figure 1. Both datasets are mostly short enough to fit inside the DNA model's context window. In addition, around 95% of codons in these datasets contain a synonym, leaving us with plenty of room for modifying a genetic sequence into a synonymous one.

As for potential challenges, both datasets contain many classes with fewer than 50 sequences. To ensure the model performs well on these smaller classes as well, we would use a metric similar to macro F1 to weight each class equally, and report uncertainty based on class size.

Another challenge is the noise in the data. After removing sequences lacking fundamental DNA elements, 2.3% of CARD and 10.6% of MegaRes entries are dropped. In addition, the largest classes are mostly different among the two datasets. A model which performs well on one may perform worse on the other. Therefore, we will be careful in reporting cross-validation and in-distribution results.

	CARD	MegaRes v2
Total nucleotide sequences	6,415	7,126
protein-coding (rRNA removed)	6,329	7,126
A/C/G/T only	6,314 (99.8%)	7,086 (99.4%)
valid protein-encoding	6,184 (97.7%)	6,372 (89.4%)
Total bases (Mb)	6.45	7.34
Antimicrobial types	1 (antibiotics)	4 in the full DB
Resistance (drug) classes	49	27
Label structure	multi-label	single-label
genes with ≥ 2 classes	64.3%	0%
Median length (nt)	873	861
Length range (nt)	162–8,673	162–5,297
Largest class (share of sequences)	54.3%	54.9%
Classes with ≥ 100 sequences	14	10
Classes with < 50 sequences	30	14
Exact duplicate sequences	26 (0.4%)	3 (0.04%)

Table 1: **Comparison of CARD and MegaRes (drugs subset).** "Valid" sequences are A/C/G/T-only, in frame (length divisible by 3), start with a start codon, end with a stop codon, and have no internal stop codon.

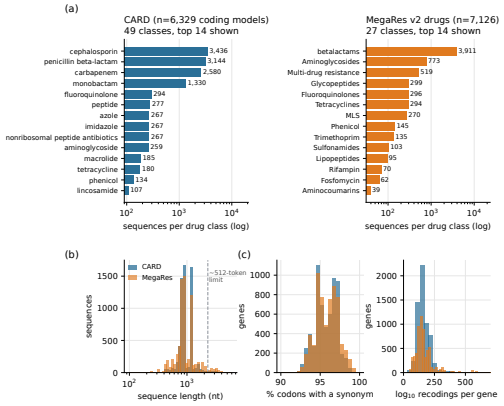


Figure 1: **Composition of the two resistance-gene datasets.** Blue: CARD; orange: MegaRes. **(a)** Number of sequences in each drug class. In CARD, 64.3% of genes are annotated with more than one class, but each gene is limited to one label in MegaRes. **(b)** Sequence lengths in nucleotides. **(c)** How much room each gene leaves for synonymous recoding.

Table 2: Summary of related work.

Work	Summary of approach
Yoo et al. [2024]	Applied substitutions to DNA sequences and found that DNA models are highly susceptible to adversarial attacks; showed that adversarial training method is a defense mechanism.
Koilakos et al. [2026]	Adversarial contamination of the training data; showed that less than 1% of poisoning nevertheless negatively impacts the model.
Kurth et al. [2026]	Modified gradient division layers of DNABERT-2 and compared it with explanable CNNs; found that gLMs provide reliable, biology-grounded explanations.
Yoo [2025]	Showed that large language models can perform similarly to genomic language models on predicting antimicrobial resistance genes.
Ma [2025]	Takes advantage of DNA sequence reversal symmetry to test model robustness. The regularization method significantly reduces prediction flips and errors.

4 Related Work

A summary of 5 prior works is shown in 2. Yoo et al. [2024] supports our motivation for this work. Kurth et al. [2026] demonstrated that Genome Language Models (gLMs) can be explained; we will reference this work if our results highlight the need for interpretability and explanation. On another aspect of adversarial robustness, Koilakos et al. [2026] investigates poisoning attacks and lights up a potential path for us to further expand our scope. Yoo [2025] showed the capabilities of LLMs in the same category of tasks which we tackle, and it would serve as a baseline.

5 Proposed Approach

We plan on intervening at the training stage by adding a contrastive term to the fine-tuning objective. Synonymous recodings of a gene should map to nearly identical representations, while frameshift and nonsense variants should be far apart.

For each gene, we build two edited copies: a positive x^+, which swaps a fraction p of codons for synonyms without changing the encoded protein, and a negative x^-, which deletes a nucleotide (frameshift) or adds a stop codon (truncation).

Contrastive objective. With the pooled gLM embedding $h_\theta(\cdot)$, we ask the model to pick x_b^+ from a lineup whose negatives $\mathcal{N}(x_b)$ contain the destructive variant x_b^- and the other $B-1$ batch genes. The destructive variant is informative because it is nearly identical as a whole sequence. With cosine similarity sim and temperature τ, the InfoNCE loss is trained jointly with the classification objective:

$$\mathcal{L}_{\mathrm{con}} = -\frac{1}{B}\sum_{b=1}^{B} \log \frac{e^{\,\mathrm{sim}(h_\theta(x_b),\, h_\theta(x_b^+))/\tau}}{e^{\,\mathrm{sim}(h_\theta(x_b),\, h_\theta(x_b^+))/\tau} + \sum_{x^- \in \mathcal{N}(x_b)} e^{\,\mathrm{sim}(h_\theta(x_b),\, h_\theta(x^-))/\tau}}, \quad \mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \lambda\, \mathcal{L}_{\mathrm{con}}. \tag{1}$$

We will select λ, which balances the objectives, by cross-validation.

Hard negative mining. Random recodings may not fool the model, so we will train a separate version that searches for troublesome recodings. Under a query budget, we will go codon by codon and keep synonymous substitutions that move embeddings the most ($\|h_\theta(x) - h_\theta(x^+)\|$), then train on those. This resembles adversarial training, but the allowed perturbations come from the genetic code.

Compared models. We will train four versions across the same data split and compute budget: a plain fine-tuned baseline; a version with perturbed training sequences, as in Yoo et al. [2024]; contrastive fine-tuning with random synonymous positives; and contrastive fine-tuning with hard-mined positives. These comparisons test whether contrastive learning improves on data augmentation and whether searching for hard cases helps. We will use DNABERT-2 [Zhou et al., 2024] and Nucleotide Transformer [Dalla-Torre et al., 2025], with LoRA to fit the compute budget.

Yoo et al. [2024] apply substitutions without examining the encoded protein, while Ma [2025] use reverse-complement symmetry, conceptually similar to our proposed method. Using the genetic code gives us protein-preserving and destructive edits, allowing us to measure both robustness and sensitivity. Our search for worst-case recodings also differs from both.

Our method only applies to protein-coding DNA, so we retain validated coding sequences (Table 1). Codon choice also reflects evolutionary preferences, and ignoring it might discard useful information. We will therefore freeze embeddings and train small probes to predict organisms of origin and GC content, comparing against the baseline. Finally, to detect representation collapse, we will monitor embedding variance and report classification performance alongside robustness.

6 Evaluation Metrics

Let $\hat{y}(x)$ be the predicted resistance class over N test genes. We will report five metrics. **Synonymous flip rate** (SFR; lower is better), $\frac{1}{N}\sum_i \mathbf{1}[\hat{y}(x_i^+) \neq \hat{y}(x_i)]$, measures how often a harmless edit changes the prediction. **Destructive flip rate** (DFR; higher is better) is the corresponding rate for frameshift/nonsense variants, where the prediction should change. **Minimum flip distance** (higher is better) is the lowest number of synonymous substitutions needed to flip a prediction, estimated by greedy search under a query budget inspired by DeepFool. **Embedding shift rate** (higher is better), $\mathbb{E}\|h(x^-) - h(x)\| / \mathbb{E}\|h(x^+) - h(x)\|$, measures the variants' relative effects on representations. Finally, **performance** will be measured by macro-F1 on unedited sequences to account for class imbalance.

The project will be considered successful if SFR drops by at least 10 points from baseline, macro-F1 stays within 2 points, and DFR does not decrease. All three conditions are needed to rule out a model that achieves consistency by ignoring its input. We also expect hard-mined positives to outperform random positives on minimum flip distance. Some items we plan to analyze include the following.

- **Performance and uncertainty:** Report all five metrics for the four configurations across both underlying models and datasets, using three seeds. Report standard deviations across seeds and bootstrap confidence intervals for per-class results, particularly for classes with fewer than 50 sequences.
- **Ablation studies:** Vary contrastive weight λ and recoding fraction p, compare random and hard positives, and remove destructive variants from the loss to test whether separating broken sequences helps.

- **Case studies:** Examine genes where a single synonymous substitution flips the baseline prediction but not our model's, and inspect the edited codon.
- **Visualization / embedding analyses:** Plot UMAP projections of genes and their variants before and after fine-tuning. Compute integrated gradients across nucleotide positions to examine whether the model weights the first two codon positions more than the third, which is usually synonymous.

7 Timeline

Table 3: Project timeline.

Date	Milestone
Oct 24	Finalize data pipeline and variant generators
Oct 27	Baseline DNABERT-2 fine-tuned on CARD with a macro-F1
Nov 3	Robustness eval code done, baseline fragility (SFR, DFR) measured
Nov 10	Contrastive objective implemented; train with random positives
Nov 14	**Project milestone:** baseline fragility and initial results
Nov 24	Mine for worst-case recodings, launch all 16 runs
Dec 1	Ablations and seed replicates complete (variance measurements)
Dec 5	Draft report
Dec 12	Case studies, plots, probing, attributions, revision
Dec 16	Final report

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